Mapping Work-Home Connections in Germany: A Network Analysis

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1 Introduction

Alongside migration, commuting represents one of the most important forms of human mobility. The topic is also of growing importance due to increasing commuter numbers [1] and a growing relevance for the topic of energy efficiency in Germany. In this context, commuter networks are usually examined in terms of distances and statistics. The use of network-based methods in commuter studies is less common in Germany. However, this approach offers great potential to open up a new perspective on the topic. Commuting networks can be interpreted in different ways depending on the application. For example, in epidemiological studies, the modelling as an undirected graph focusing on the general connections between territorial units is of interest. This is because commuters, unlike migrants, complete both the outbound and return journeys in each case. If the graph model of commuter connections is considered as directed, meaning that the difference between the workplace and residence is included in the analyses, the analyses would, on the contrary, be relevant, for example, for socioeconomic structural analyses.

Due to this advantage for commuter relationships, a directed graph is created in this article based on commuter data from the Federal Employment Agency in Germany. In this graph, the counties of Germany represent the nodes and the commuter connections between the counties represent the edges of the graph. The commuter data of the Federal Employment Agency, BA (Bundesagentur für Arbeit) (dataset "Arbeitsmarkt in Zahlen, Sozialversicherungspflichtig Beschäftigte - Pendler nach Kreisen") contains the commuter flows between counties as well as connections abroad since 2013 and is published annually. Connections abroad were not taken into account in this work. Technical problems or administrative reorganizations in 2013/14, 2016/2017, and 2019 should be mentioned in connection with data quality. In addition, when interpreting the results, especially before the increasing importance of the topic of home office, it must be taken into account that the data set is derived from employment contracts and do not necessarily represent actual trips, such as mobile phone data. For georeferencing of territorial units, the Federal Agency for Cartography and Geodesy (BKG) now also provides historical data. For the georeferencing of the commuter data, the dataset Verwaltungsgebiete 1 : 250 000 (VG250-EW) (© GeoBasis-DE / BKG respective year of cut-off date) was used according to the temporal correspondence.

2 Results

Initial analyses match the results of the Federal Office for Building and Regional Planning - BBSR, which show an increase in commuter numbers, especially among commuters with a commute of more than 50km.[1] From the commuter data, directional graphs were created for the respective years from 2013 to 2022. Due to the change in the county structure, the number of nodes (counties) decreases from 402 in 2013 to 400 in 2022, while the number of edges increases from 38631 in 2013 to 47082 in 2022. The maximum in-degree values are at (or since 2017 permanently on) the upper boundary, i.e. there is at least one hub that attracts commuters from all other counties in Germany. Looking at the out-degree values, these are also close to the possible maximum, but below the in-degree values. Since the out-degree values are below the maximum, an increase can also be seen here. The minimum degree values fluctuate more, with the minimum out-degree values above the minimum in-degree values. This is probably due to structurally weak counties, which are themselves of little interest as places to work. Fittingly, the median of the out-degree values is also higher than the median of the in-degree values. However, both show a similar increase over time.
When analyzing the community structure using the Rb Pots model [2], the commuter network shows a community structure (Figure 1) that matches very well with existing metropolitan regions in Germany. A change in community membership over the years occurs particularly in structurally weaker regions such as Saxony-Anhalt.

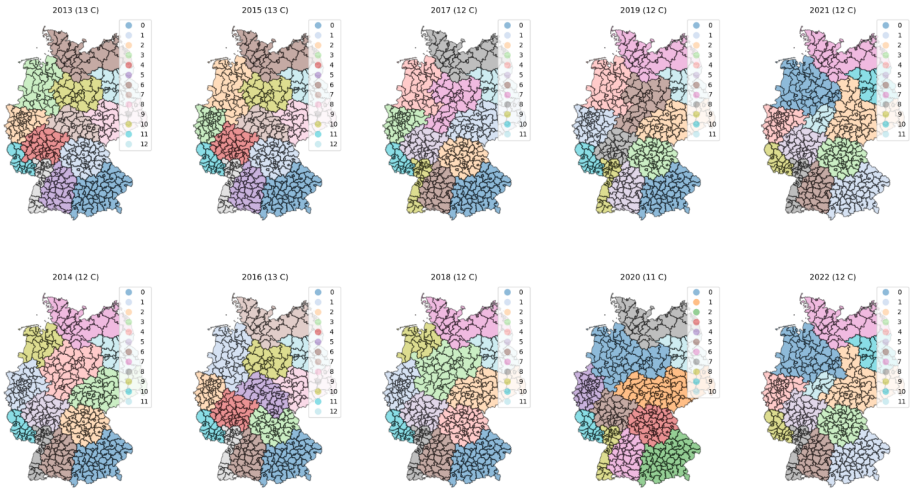

Fig. 1. Community structure

Community 9 (2013) in the Karlsruhe-Mannheim area is interesting here, extending in a southerly direction via Freiburg to the state border. To the west, the state border limits the expansion, but to the east of the community is Stuttgart, a nearby metropolitan region. The reason for the expansion is probably the course of the A5 highway, which

runs in a north-south direction from Basel to Darmstadt, parallel to the Rhine through the community. An analysis of the in- and out-degree centrality also shows higher median values for the out-degree than for the in-degree. The values also increase over the period under study. The distribution of counties is interesting here. Among the ten territorial units with the highest in-degree centrality, both in 2013 and in 2022, there are exclusively populous territorial units, predominantly independent cities (7 out of 10) such as Munich, Berlin, Frankfurt am Main or Hamburg. The only two counties (Munich and Hannover Region) in the top 10 are also very populous. This also applies to the only urban district on the list (Stuttgart). In 2022, the top three places Munich, Berlin, and Frankfurt am Main have an in-degree centrality of 1, meaning they have commuters from all other nodes. Berlin is also the only territorial unit of the new federal states in the top 10, even though Berlin plays a special role in this context.
The analysis of out-degree centrality gives similar results. However, in 2022, Leipzig and Dresden, two further territorial units of the new federal states, are on the list of the top 10. Not among the top 10 out-degree in 2022 are the district of Munich and the independent city of Düsseldorf. In 2013, the Erzgebirgskreis, another district of the new federal states, is on the list. It is also interesting to note the different ranges for the top 10 in In-Degree, between 1 and 0.91, and in Out-Degree, between 0.99 and 0.63.

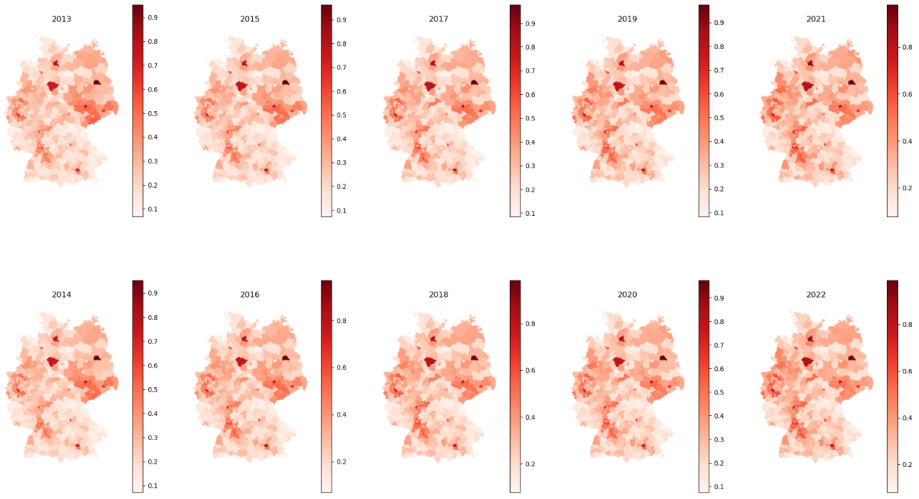

Fig. 2. Out-Degree-Centrality over period

Among the last ten places in the in-degree centrality rankings, Kaufbeuren in 2013 and Suhl in 2022, are the only district-free cities. In contrast rankings of the ten counties with the lowest out-degree centrality are dominated by district-free cities (10 out of 10 in 2013, 7 out of 10 in 2022), such as Weiden i. d. Oberpfalz or Amberg. This may be due to a small population in combination with an attractive labor market and less economically attractive surrounding regions.

Summary. The results of the network-based analysis of commuting data for Germany are in line with the results of classical statistical and spatial analyses, but offer an additional perspective on commuting. The use of a directed graph allows to distinguish between place of work and place of residence.

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